scores formed

weights aggregated

written, and cost of the aggregation

Dense, unfused
writes an $N \times N$ matrix
$2N^2d$

Dense, fused
writes no intermediate
$2N^2d$

Top-k, masked
writes an $N \times N$ matrix
$2N^2d$

Top-k, gathered
writes an $N \times k$ tensor
$N^2d + Nkd$

Top-k, fused
writes no intermediate
$N^2d + Nkd$

A dashed outline marks weights that are used but never written to memory; a cross marks the masked self-interaction.